

Task 6: Final Project Report

1. Problem Definition

The project predicts whether an adult earns more than \$50K to support prioritization of potential high-income prospects.

2. Data Preparation

OpenML Adult version 2 was cleaned by mapping the target to 0/1 and replacing `?` with missing values. Age buckets, part-time status, capital-gain indicators and log transform, higher-education, overtime, and married-status features were reproduced deterministically. A stratified 80% training-development and 20% untouched test split was used, with a further development split for threshold selection. The production wrapper now applies feature engineering automatically before the saved model.

3. Model Development

Logistic Regression, Random Forest, and Gradient Boosting were evaluated. On the test set at threshold 0.50, Logistic Regression achieved F1 `0.6754` and Gradient Boosting achieved F1 `0.6839`. The calibrated final Logistic Regression model was selected because its development threshold optimization produced the strongest final F1 for the chosen business objective.

4. Model Tuning

Day 4 used reproducible 5-fold StratifiedKFold randomized searches with F1 scoring. The selected Logistic Regression configuration was L1 regularization with C=3 and the liblinear solver. The classification threshold was selected separately on the development set as `0.30`.

5. Diagnostics

Learning curves showed a small train-validation gap that narrowed as training size increased, with both curves flattening near F1 `0.67`. Sigmoid calibration improved the test Brier score slightly from `0.0978` to `0.0977`. Threshold selection increased recall and F1 while reducing precision and accuracy relative to threshold 0.50.

6. Final Results

At threshold `0.30`, the final test metrics were Accuracy `0.8367`, Precision `0.6241`, Recall `0.7990`, F1 `0.7008`, ROC-AUC `0.9135`, PR-AUC `0.7869`, and Brier score `0.0977`. The confusion matrix was TN `6306`, FP `1125`, FN `470`, TP `1868`.

7. Model Interpretation

The largest positive coefficient was `Log\_CapitalGain`; large negative or positive effects also appeared for capital-gain presence, marital status, age, occupation, country, and sex indicators. The opposing capital-gain coefficients should be reviewed for redundancy and stability. Sex and other demographic indicators may act as historical proxies, so coefficients should not be interpreted causally.

8. Production Readiness

The original Day 4 calibrated artifact was loaded without retraining. A production wrapper containing the existing feature transformation and fitted model was saved as `final\_adult\_income\_production.joblib`. The inference workflow loads this wrapper, accepts raw Adult rows, returns probabilities, applies threshold `0.30`, and was tested on 10 unseen examples. Recreated split indices confirmed no overlap among train, development, and test data.

9. Limitations and Future Improvements

Performance is weaker for younger adults, lower-education groups, and females, so subgroup recall and calibration should be monitored. Future work should validate an explicit cost-sensitive threshold, evaluate newer representative data, test coefficient stability, monitor drift, review proxy-sensitive features, and add input-schema validation around the serving wrapper.